



# Git

## *Introduction*

Imagine you are working on a **group project** with your classmates. Everyone writes code, but someone **accidentally deletes an important file**. How do you get it back? Or what if you make changes but later realize the old version was better?

This is where **Git helps**.

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## *Section 1: Learn*

### What is Git?

**Git is a version control system (VCS)** that tracks changes in code, allowing multiple developers to work together **without overwriting each other's work**.

It helps:

- **Save different versions** of your code.
- **Work collaboratively** with a team.
- **Go back to an earlier version** if needed.

### Why is Git Important?

1. **Prevents data loss** – Keeps a history of code changes.
2. **Collaboration** – Multiple people can work on the same project.
3. **Reverts mistakes easily** – If something breaks, you can go back to a previous version.
4. **Works offline** – Unlike cloud-based tools, Git works on your local computer.



5. **Industry Standard** – Used by companies like **Google, Microsoft, Amazon, and Flipkart**.

## How Does Git Work?

Git follows a simple process:

1. **Initialize a Repository** – Start tracking a project with Git.
2. **Stage Changes** – Select files to include in the next commit.
3. **Commit Changes** – Save the current state of the code.
4. **Push to Remote** – Upload changes to a cloud repository like **GitHub**.
5. **Pull Updates** – Download the latest code from the cloud.

## An Interesting Anecdote

Before Git, developers used **manual file backups** or emailed files to each other. This led to **lost changes and confusion**. In 2005, **Linus Torvalds (the creator of Linux)** developed Git to manage **thousands of code changes** efficiently. Today, **Git is the most widely used version control system worldwide**.

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## Section 2: Practice

### Step-by-Step Guide to Using Git

#### 1. Setting Up Git

Before using Git, you need to **install and configure it**.

```
# Check if Git is installed
```

```
git --version
```

```
# Set up your user name and email
```

```
git config --global user.name "Your Name"
```



```
git config --global user.email "your.email@example.com"
```

### Why is this important?

- Ensures **every commit has your name attached**.
  - Helps in tracking **who made which changes**.
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## 2. Initializing a Git Repository

```
# Create a new project folder and navigate into it
```

```
mkdir my_project
```

```
cd my_project
```

```
# Initialize Git in the folder
```

```
git init
```

### What happens here?

- Git starts **tracking all changes** in this folder.
  - Creates a **hidden .git folder** to store history.
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## 3. Adding and Committing Changes

```
# Create a new file
```

```
echo "print('Hello, Git!')" > script.py
```

```
# Add the file to Git staging
```

```
git add script.py
```

```
# Commit the changes
```



```
git commit -m "Added script.py file"
```

What does this do?

- **git add** prepares the file for commit.
  - **git commit** saves the file with a message describing the change.
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#### 4. Pushing to GitHub (Remote Repository)

To save your code online, you need a **GitHub repository**.

```
# Connect local Git repository to GitHub
git remote add origin https://github.com/yourusername/my_project.git

# Push code to GitHub
git push -u origin main
```

Why is this useful?

- Keeps a **backup** of your code online.
  - Allows your **team to access and collaborate** on the project.
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#### 5. Pulling Changes from GitHub

If someone else has updated the code, you need to **sync your local copy**.

```
# Pull the latest changes
git pull origin main
```

How does this help?

- Updates your **local copy** with the latest changes.
  - Prevents **conflicts between team members**.
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## ***Real-World Example***

A **software company** is developing a **mobile banking app**. Their **10-member team** works on different features like **login, payments, and security**.

### **Without Git:**

- Team members **overwrite each other's work**.
- It is **hard to track who changed what**.
- If a bug appears, there is **no easy way to roll back**.

### **With Git:**

1. Each team member **works on a separate branch**.
2. Changes are **tracked and saved** automatically.
3. If something goes wrong, they can **roll back to a stable version**.

**Result?** Faster development, **fewer errors**, and **better collaboration**.

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## ***Section 3: Know More***

### **Frequently Asked Questions**

#### **1. Is Git the same as GitHub?**

No. **Git** is a **version control tool** used locally, while **GitHub** is an **online platform** to store and share Git repositories.

#### **2. Can I use Git without an internet connection?**

Yes! Git works **offline**, but you need the internet to **push and pull changes from GitHub**.

#### **3. How do I undo changes in Git?**

```
# Undo the last commit (without deleting the changes)
```

```
git reset --soft HEAD~1
```



```
# Undo the last commit and delete the changes
```

```
git reset --hard HEAD~1
```

#### 4. What is the difference between Git and SVN?

- **Git** is **distributed**, meaning every developer has a full copy of the repository.
- **SVN** is **centralized**, meaning there is only one central repository.

#### 5. Where can I practice Git?

- Create a **GitHub account** and start a sample project.
- Try **committing and pushing changes** using Git.

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### **Conclusion**

Git is an **essential tool for every developer**. It helps teams **collaborate, track changes, and work efficiently**. By mastering Git, you can **manage code better and work on real-world projects smoothly**.

Start with **basic Git commands**, explore **GitHub**, and soon you'll be contributing to **big projects confidently**!